

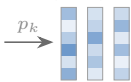
(a) A learned connection is a paired memory

forward history $h_a^{(\ell-1)}$



clock $\xi \in [0, \tau]$

$\bar{h}_{a,0} \bar{h}_{a,1} \bar{h}_{a,2}$

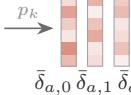


fixed basis p_k ,
moving n -vectors

backward history $r_a \delta_a^{(\ell)} / \rho$



clock $\xi \in [0, \tau]$

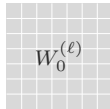


$\bar{\delta}_{a,0} \bar{\delta}_{a,1} \bar{\delta}_{a,2}$



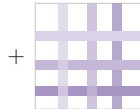
pair

$$\sum_{a,k} (2k+1) \bar{\delta}_{a,k} \bar{h}_{a,k}^\top$$



kept, fixed

$\widehat{W}^{(\ell)}$



rank $\leq mq$

(b) Why pairing helps

$\int b h^\top$ splits into four pairings

	$\Pi_q h$	$h - \Pi_q h$
$\Pi_q b$	kept memory	0
$b - \Pi_q b$	0	defect omitted $\times$ omitted

mixed terms vanish by orthogonality

feedback
stability

trajectory error $O_T(q^{-1})$ or $O_T(q^{-2})$

responses of the reconstructed network feed the next memories